

Dataset Overview

Aadi Ojha - Bethlehem Central High School

Sources

- <https://www.wi-lab.net/datasets-page/> - Arizona State University
- <https://www.rfdatafactory.com/datasets> - Northeastern University

Arizona State University - Wireless Intelligence Lab

Breakdown

- 39 different data sets; can be sorted into 4 categories based on the type of scenario and its application

Category	Application	Datasets
Vehicle to Infrastructure	Beam prediction	1,2,3,4,5,6,7,8,9,13,14,15,31,32,33,34
Fixed Tx/Rx street level communication	Blockage prediction	17,18,19,20,21,22,24,25,26,27,28,29,30
Human Data Collection	Beam/blockage prediction	10,11,12
Vehicle to Vehicle	Beam Prediction	36,37,38,39

- Scenario 16 - Indoor communication - blockage prediction
- Scenario 23 - mmWave communication - beam prediction

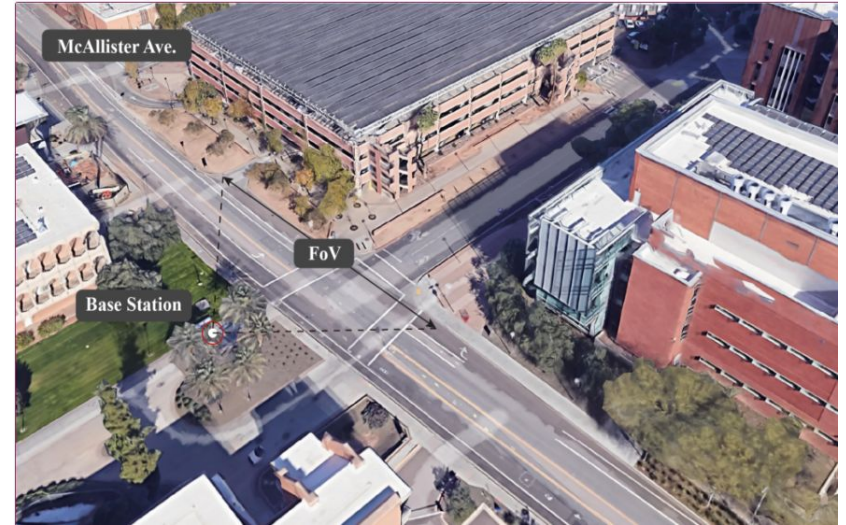
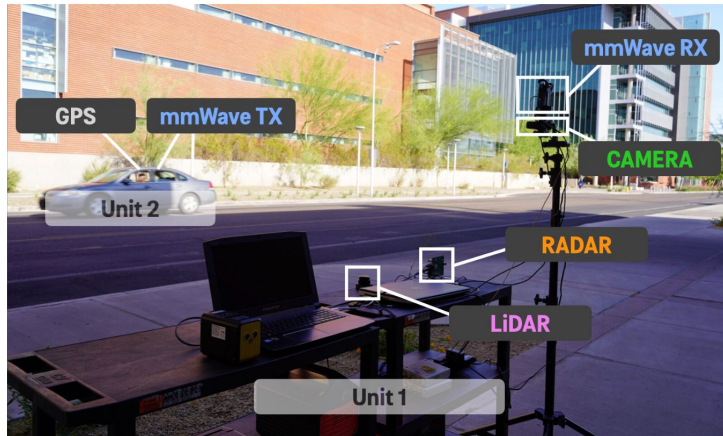
Vehicle to Infrastructure Example - Dataset 1

- Designed to test mm wave communication
- Unit 1 - Stationary base with:
 - RGB camera
 - mmWave phased array
 - 16 element 60GHz-band phased array
 - 64 predefined waves
- Unit 2 - Moving vehicle with:
 - mmWave transmitter
 - GPS
 - Quasi-omni antenna constantly transmitting at 60GHz-band



Vehicle to Infrastructure Example - Dataset 1

- Experimental Setup:
 - Two way street with 2 lanes
 - Width - 10.6 meters
 - Vehicle Speed Limit - 25mph
 - Includes walking and cycling traffic



*Note the use of LiDAR and RADAR. Some datasets include this data as well

Hardware Elements: Specifications

Unit 1:

- RGB Camera
 - 1100 FoV – 30 FPS – Wide Angle – f/1.8 Aperture
 - Output: RGB images of dimensions 960 x 540
- mmWave receiver
 - 60GHz phased array – 16-element ULA – 64-beam codebook
- Radar:
 - FMCW radar with 3 Tx & 4 Rx antennas – fully-digital reception– 10 FPS – frequency range 76- to 81 GHz with 4 GHz bandwidth – Max. range ~ 100 m – Range resolution of 60 cm
 - Output: 3D complex I/Q radar measurements of (number of Rx antenna) x (samples per chirp) x (chirps per frame) : 4x256x128
- Lidar
 - Range: 40 m – 10 FPS – Resolution of 3 cm – FoV of 3600 – Accuracy ~ 5 cm – Wavelength ~ 905 nm
 - Output: 360-degree point cloud sampling data

Unit 2

- GPS:
 - 10 measurements per second – Horizontal positional accuracy ~ 2.5m without RTK fix and ~ 10cm with RTK fix
 - output: Latitude and Longitude among other values
- mmWave Transmitter
 - 60GHz – Omni transmission (always oriented towards the receiver)

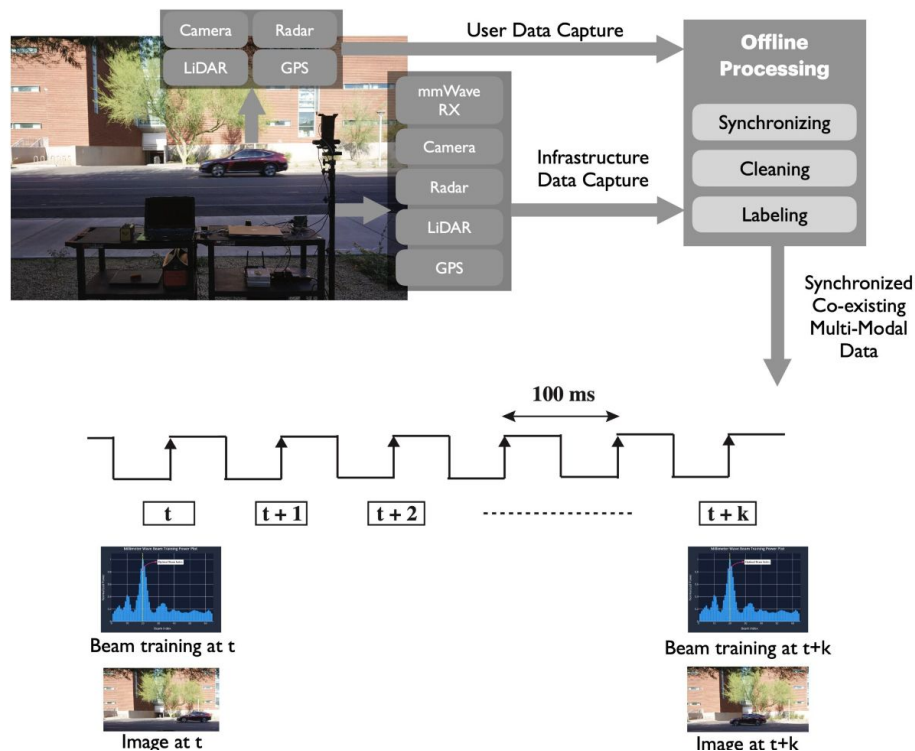
Data Collection

Visualization



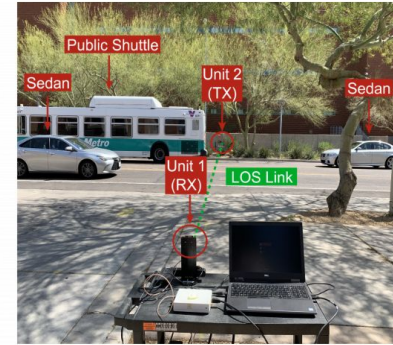
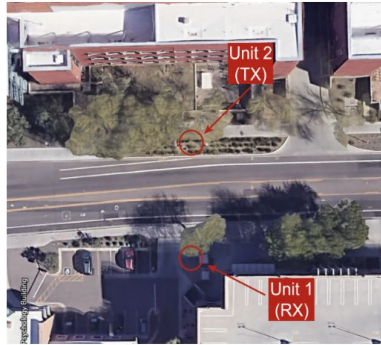
Pipeline

Data Collection Process



Fixed Tx/Rx Street Level Communication - Dataset 17

- Measuring **blockage**
- Two units on alternate sides of a two way street
- Transmitter - Unit 2
 - Emits beams using one antenna (mmWave)
- Receiver - Unit 1
 - Continuously scanning for beams
 - Uses a steering codebook of 64 beams
 - Measures the power of each beam
 - Collects data samples at a dynamic rate
 - 6-12 seconds
 - Average - 10 sec.



Fixed Tx/Rx Street Level Communication - Dataset 17

- Experimental Setup
 - Two way street with 2 lanes
 - Width - 10.6 meters
 - Vehicle Speed Limit - 25mph
 - Includes blockage from other vehicles - diverse data set



Hardware Elements: Specifications

Unit 1:

- RGB Camera
 - 1100 FoV – 30 FPS – Wide Angle – f/1.8 Aperture
 - Output: RGB images of dimensions 960 x 540
- mmWave receiver
 - 60GHz phased array – 16-element ULA – 64-beam codebook

Unit 2

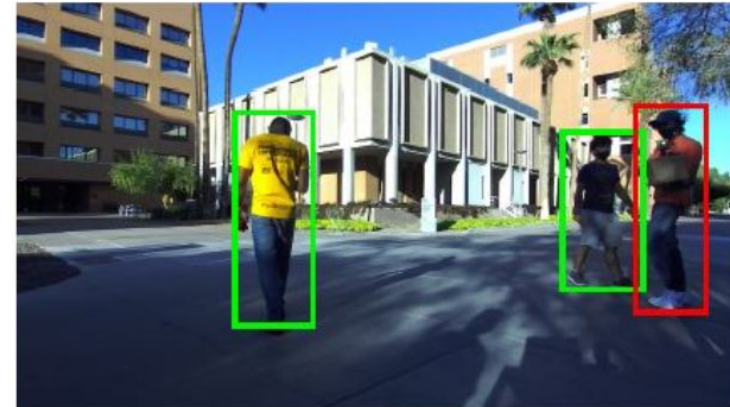
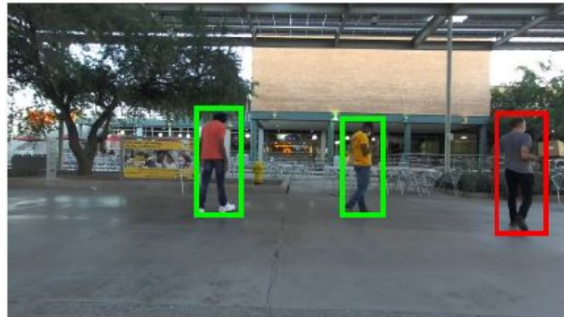
- mmWave Transmitter
 - 60GHz – Omni transmission (always oriented towards the receiver)

Data Visualization



Human Data Collection - Dataset 10

- Collecting data with moving humans
- Unit 1 - Stationary base with:
 - 16 element 60 GHz phased array
- Unit 2 - Mobile person with:
 - mmWave transmitter with quasi-omni antenna
 - Transmitting 60 GHz band
 - GPS to track location



Hardware Elements: Specifications

Unit 1:

- RGB Camera
 - 1100 FoV – 30 FPS – Wide Angle – f/1.8 Aperture
 - Output: RGB images of dimensions 960 x 540
- mmWave receiver
 - 60GHz phased array – 16-element ULA – 64-beam codebook
- Radar:
 - FMCW radar with 3 Tx & 4 Rx antennas – fully-digital reception– 10 FPS – frequency range 76- to 81 GHz with 4 GHz bandwidth – Max. range ~ 100 m – Range resolution of 60 cm
 - Output: 3D complex I/Q radar measurements of (number of Rx antenna) x (samples per chirp) x (chirps per frame) : 4x256x128
- 3D Lidar
 - Resolution: 32 Channels (Vertical) and 1024 (Horizontal) – Range: 120 m – FPS: 10/20 – FoV : 45° ($\pm 22.5^\circ$) Vertical and 360° Horizontal – Accuracy ~ 5 cm – Wavelength: 865 nm
 - Output: 360-degree point cloud sampling data

Unit 2

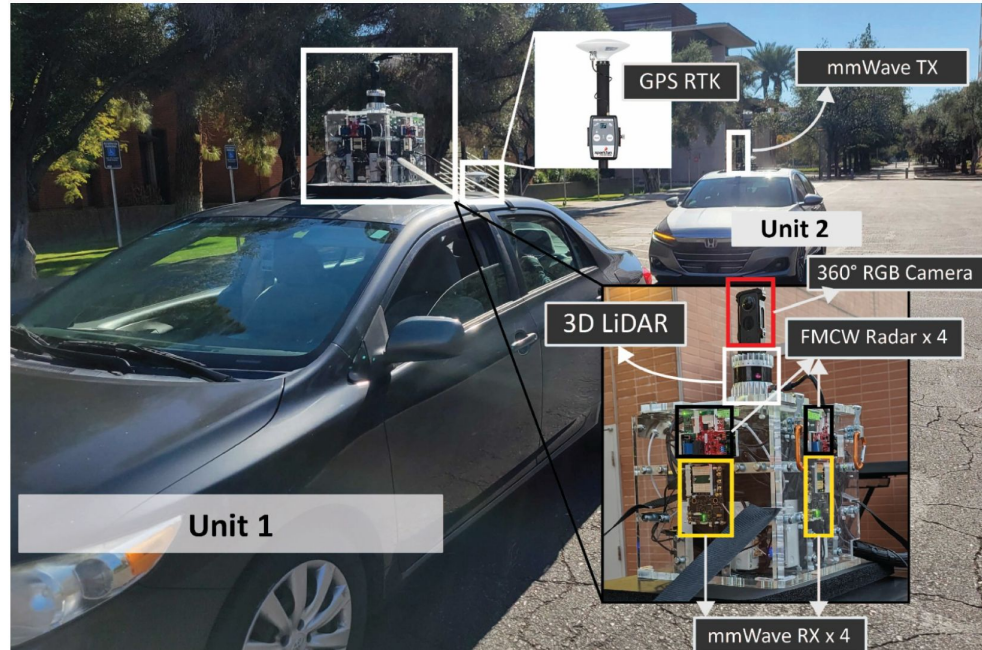
- GPS-RTK with GNSS Multi-Band L1/L2 Surveying Antenna:
 - 10 measurements per second – Horizontal positional accuracy ~ 2.5m without RTK fix and ~ 10cm with RTK fix
 - output: Latitude and Longitude among other values
- mmWave Transmitter
 - 60GHz – Omni transmission (always oriented towards the receiver)

Data Visualizations

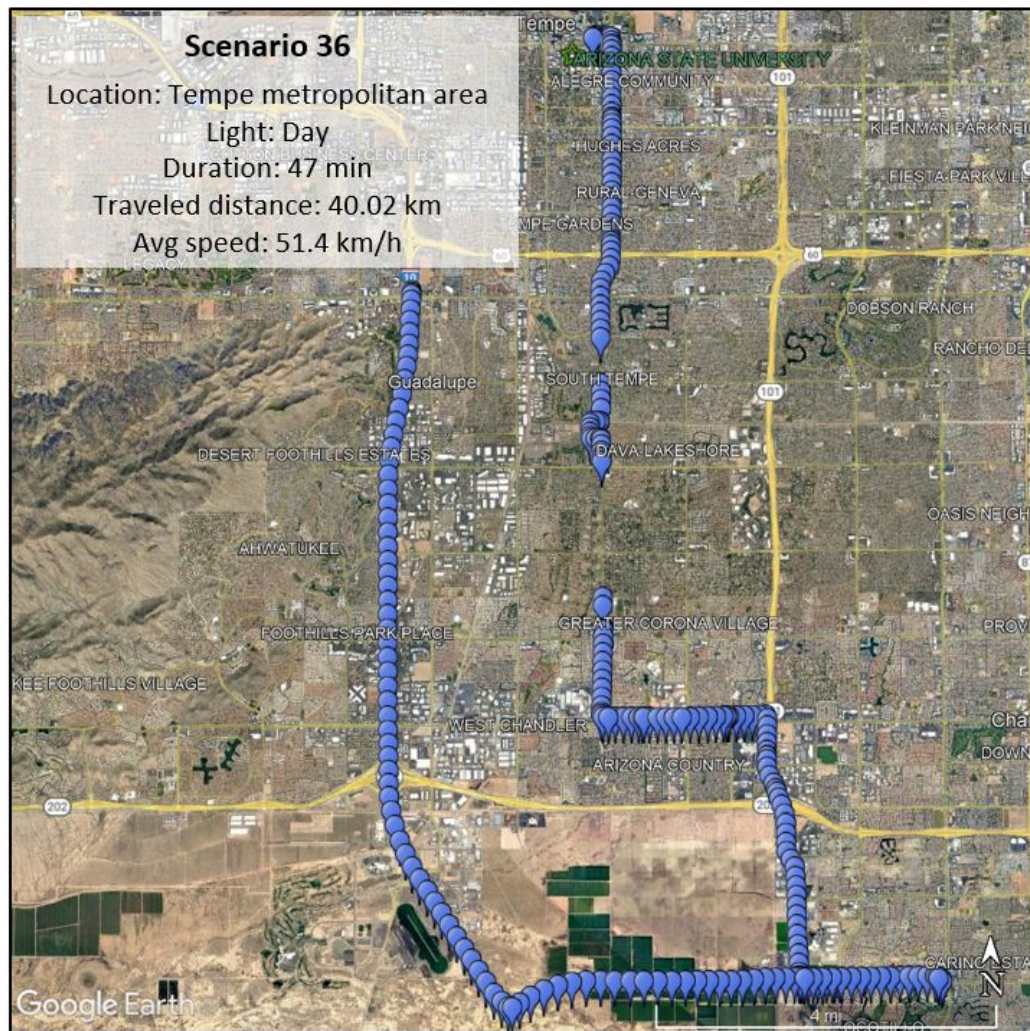


Vehicle to Vehicle - Dataset 36

- Measures beam power between vehicle to vehicle communication
- Unit 1 - Mobile Receiver with
 - 4 60GHz phased arrays facing the 4 cardinal directions
 - Each phased array uses a uniform linear array with 16 elements and a predefined codebook of 64 beam directions
 - 360 degree RGB camera
 - 4 mmWave single chip FMCW radar
 - 3D Lidar - 32 vertical and 1024 1024 azimuth channels
 - GPS RTK Kit
- Unit 2 - Mobile transmitter with
 - Quasi-omni antenna
 - GPS



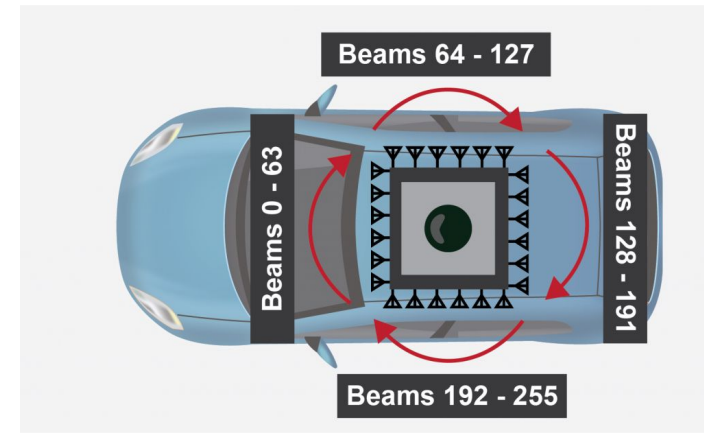
Map



Hardware Elements: Specifications

Unit 1:

- 360 RGB Camera
 - 360° 5.7K resolution @ 30 FPS
 - Aperture: F2.0
 - Focal length (35mm equivalent): 7.2 mm
- mmWave receiver (4x Phased array)
 - Each phased array is a 16-element ULA operating at 62.64 GHz center frequency using a predefined 64-beam codebook that sweeps from -45° to +45° in azimuth (approximately)
- GPS-RTK
 - 10 measurements per second – Horizontal positional accuracy ~0.5 m without RTK fix and ~ 1 cm with RTK fix
 - Output: Latitude and Longitude among other values
- Radar:
 - FMCW radar with 3 Tx & 4 Rx antennas – fully-digital reception– 10 FPS – frequency range 76- to 81 GHz with 4 GHz bandwidth – Max. range ~ 100 m – Range resolution of 60 cm
 - Output: 3D complex I/Q radar measurements of (number of Rx antenna) x (samples per chirp) x (chirps per frame) : 4x256x128
- 3D Lidar
 - Resolution: 32 Channels (Vertical) and 1024 (Horizontal) – Range: 120 m – FPS: 10/20 – FoV : 45° ($\pm 22.5^\circ$) Vertical and 360° Horizontal – Accuracy ~ 5 cm – Wavelength: 865 nm
 - Output: 360-degree point cloud sampling data
- Unit 2
 - Same as previous scenario

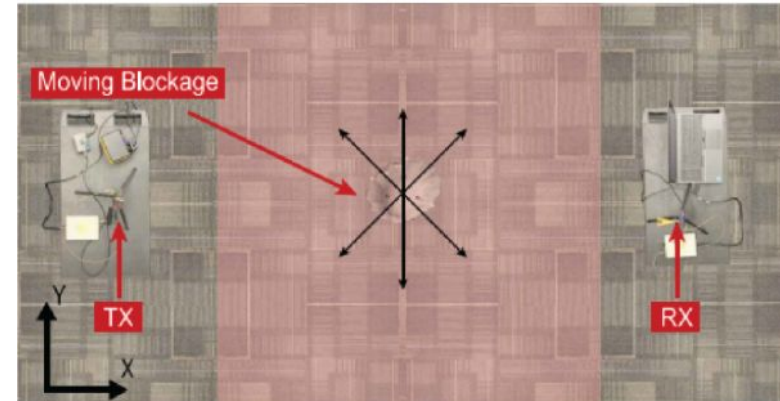


Data visualization



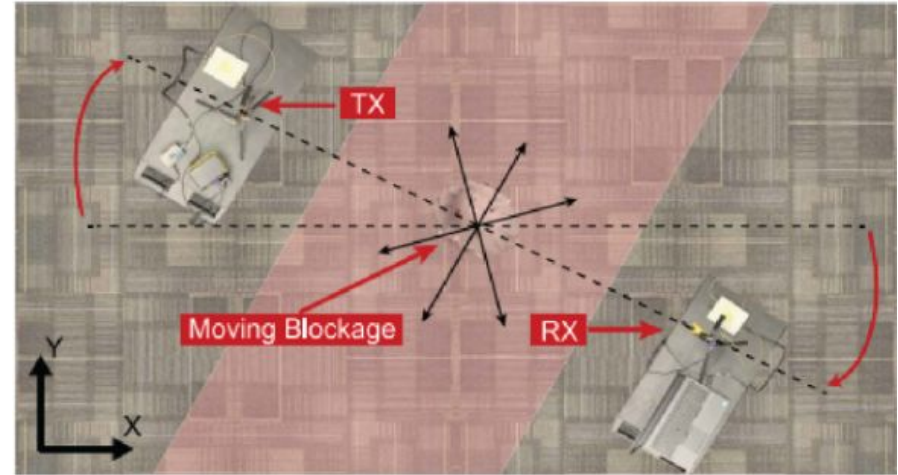
Indoor Communication - Scenario 16

- Designed to study indoor blockage
- Two stationary units, Unit 1 and Unit 2
- Unit 1:
 - 60 GHz mmWave receiver
 - 10 degree beamwidth (20 dBi) horn antenna
 - RGB antenna
- Unit 2:
 - 60 GHz mmWave transmitter
 - Omni-directional antenna
 - Power limited to 30 dBm
- Signals controlled by a computer which pass through two 2901 NI USRPs
- Coordinate data collections using IEEE 802.11 control/communication channel
- Transmit/receive waveform adopts an OFDM structure with 20MHz bandwidth and 64 carriers



Indoor Communication - Scenario 16

- Metal blockage in the form of a cylinder in the middle of the two (transmitter and receiver)
- Tx and Rx face each other with LOS connection with a distance of about 8 meters between them
- Blockage moves in two straight lined trajectories 1m apart
 - Moves at 0.0625 m/s
- Testbed collects samples at 1.13 samples/second



Hardware Specifications

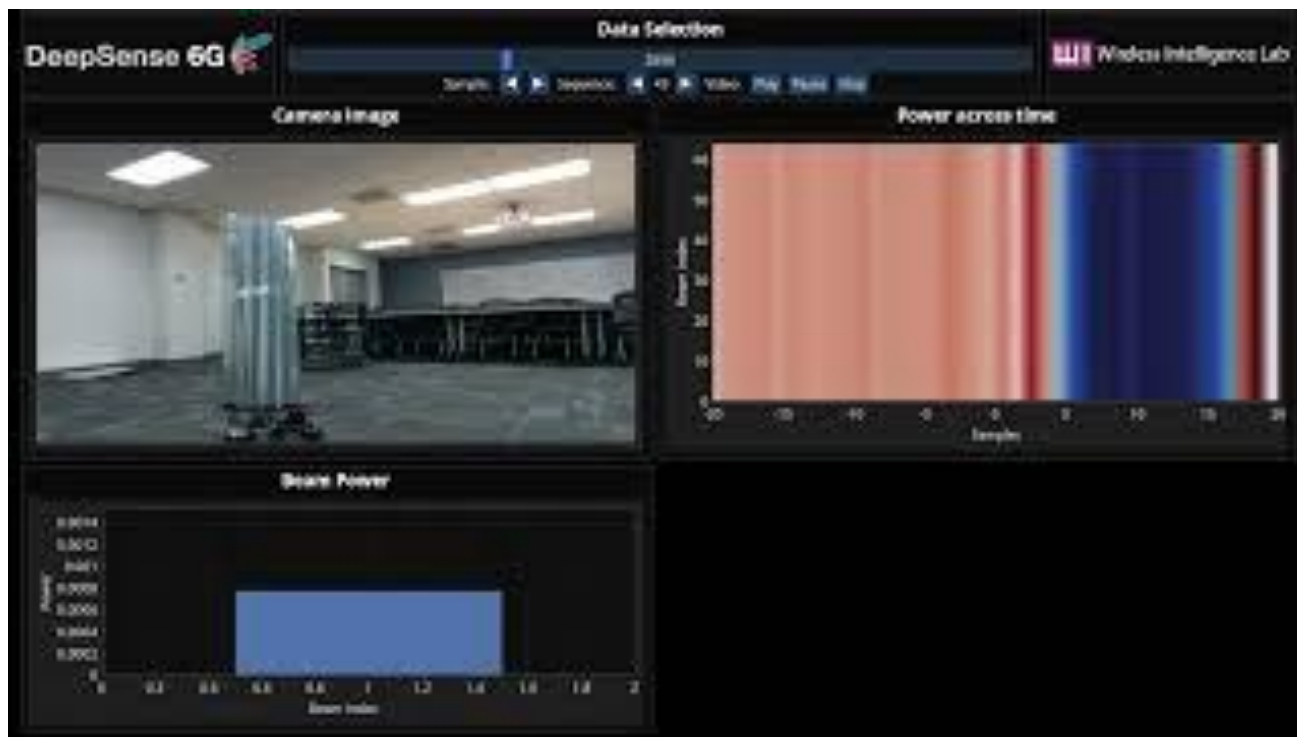
Unit 1:

- RGB Camera
 - 69° x 42° (H x V) FoV – 30 FPS – Rolling shutter – 2MP sensor resolution
 - Output: RGB images of dimensions 1920 x 1080
- mmWave Receiver
 - 60GHz – 10° beamwidth (20 dBi) horn antenna

Unit 2:

- mmWave transmitter
 - 60GHz – Omni transmission (always oriented towards the receiver) – 20MHz bandwidth – 64 subcarriers

Data Visualization



Drone Communication - Scenario 23

- Unit 1:
 - Stationary base with:
 - RGB Camera
 - mmWave Phased Array
 - 16 element 60GHz phased array of 64 predefined waves
 - Placed on table 1.5 meters above the ground facing towards the sky
- Unit 2:
 - RC Drone with:
 - mmWave transmitter
 - GPS Receiver
 - Inertial Measurement Units
 - Quasi omni antenna transmitting at 60 GHz band



Location Details

- Rectangular Park with perimeter of 1.24 km
- RC Drone flying area is 205 meters by 105 meters

Hardware Specifications

Unit 1:

- RGB Camera
 - 110° FoV – 30 FPS – Wide Angle – f/1.8 Aperture
 - Output: RGB images of dimensions 960 x 540
- mmWave Receiver
 - 60GHz phased array – 16-element ULA – 64-beam codebook

Unit 2:

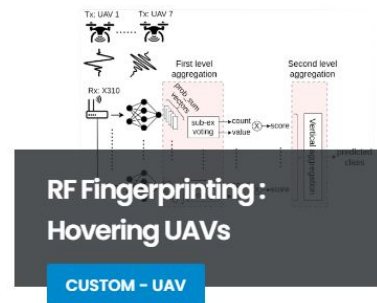
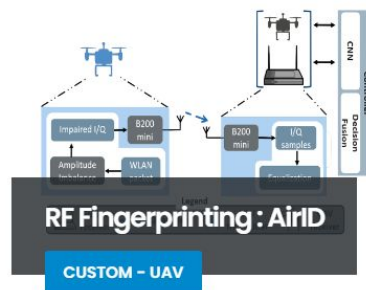
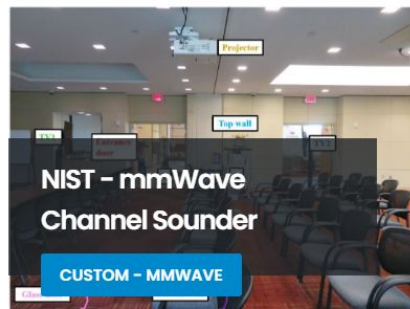
- mmWave Transmitter
 - 60GHz – Omni transmission (always oriented towards the receiver)
- GPS: A3 Flight Controller
 - Hovering accuracy: Horizontal ~ 1.5m and Vertical ~ 0.5m – D-RTK GNSS: Horizontal ~ 0.01m and Vertical ~ 0.02m
 - output: Latitude and Longitude, among other values

Data Visualization



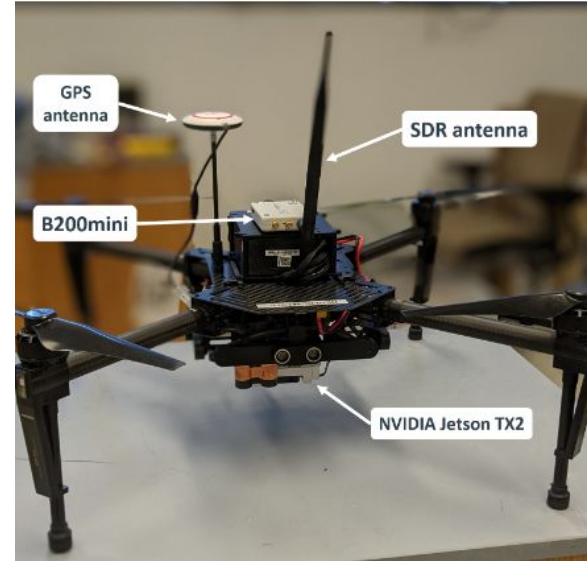
Northeastern University Datasets

Datasets in Question



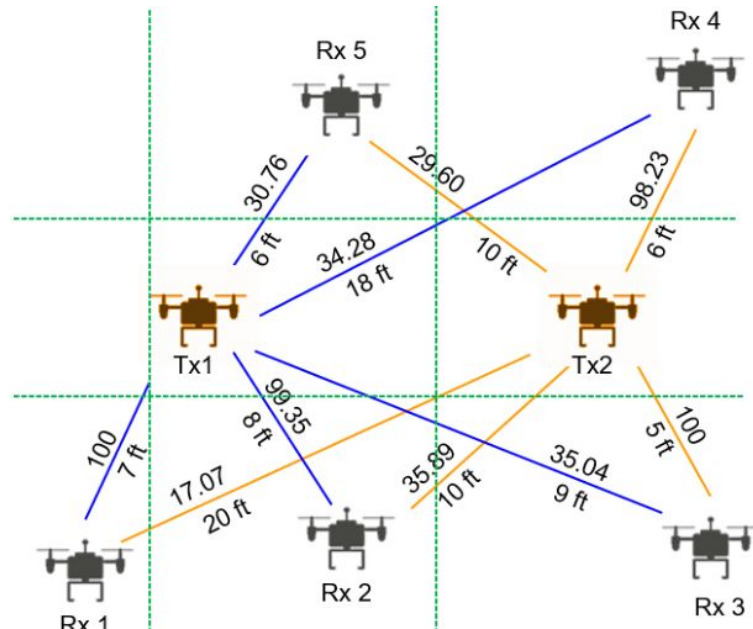
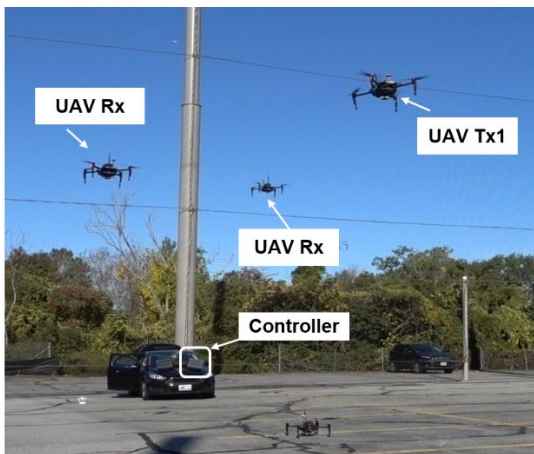
Raw UAV Transmissions - AirID

- Purpose? - These datasets were created to test a machine learning model called AirID designed to identify unmanned aerial aircrafts (UAVs)
- Setup:
 - 2 UAV transmitters (Tx) and 5 UAV receivers (Rx)
 - Tx1 and Tx2 transmit data with predefined amplitude impairments of 4 dB and 2 dB
 - Allows for unique identification
 - Rx 1, 2, and 5 are positioned close to Tx1
 - Rx 3 and 4 are positioned close to Tx2
 - Rx5 receives external interference of -10dB
 - Rx3 is performs unstable hovering when it is collecting data from only Tx1



More descriptions

- Dataset includes raw I/Q samples
- Over the air transmission of 4 USRP B200 mini radios
- UAV - DJI M100 UAV
 - Each transmits at a unique IQ imbalance



RF Fingerprinting: Hovering UAVs

- RF Fingerprinting - discriminating transmitter generated features at the receiver
 - i.e I/Q phase imbalance
 - Going to use raw I/Q samples for RF fingerprinting the hovering UAVs
- Set-up
 - 7 DJI M100 UAVs as transmitters
 - Ettus USRP X310 used to receive the signals
 - Fly UAVs one at a time at different distances from the receiver (6, 9, 12, 15ft)
 - Receiver collects I/Q samples only on the downlink 10MHz channel
 - Every 2 seconds with 10 second pause
 - During 10 second pause, signals are partitioned into 4 non-overlapping bursts each containing about 140 interleaved short periods of data and noise
 - Extract data from noise which results in about 140 sequences per burst, referred to as examples

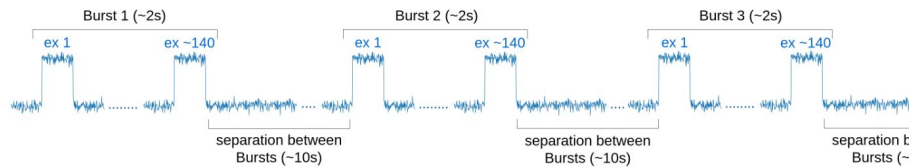


Figure 2. An overview of a sample signal collected from a hovering UAV at a certain distance.

- 13k examples with 92k I/Q samples in the dataset